nature portfolio

https://doi.org/10.1038/s41586-025-08963-7

Supplementary information

A complementary two-dimensional material-based one instruction set computer

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Supplementary Information

One instruction set computer built from complementary two-dimensional materials

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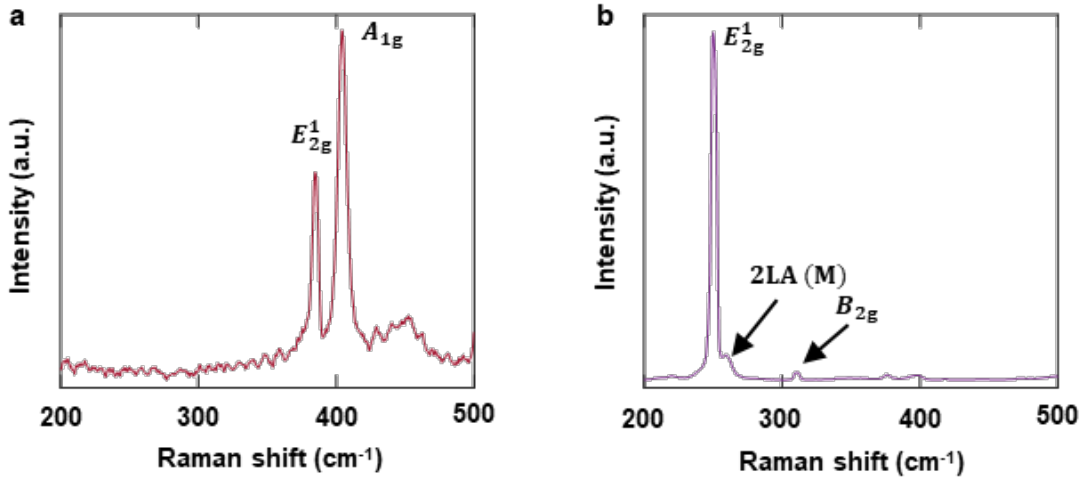
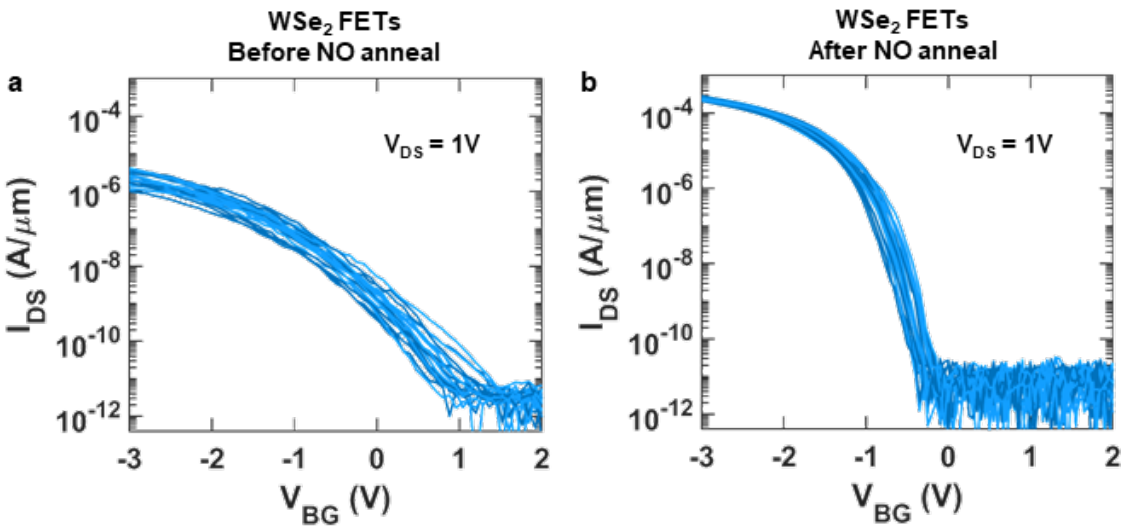
Supplementary Figure 1



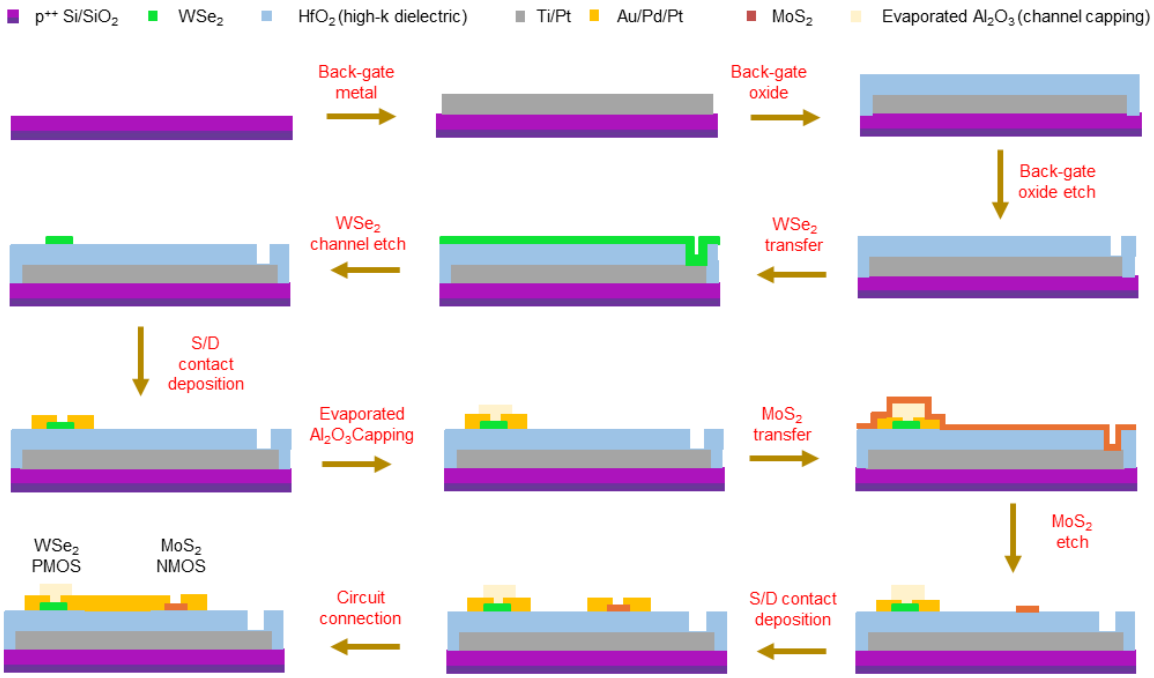
Supplementary Fig. 1. *Raman spectra for* ***a)*** *MoS_2 and* ***b)*** *WSe_2 films obtained using a 532 nm laser. The Raman active vibrational modes in the in-plane E_{2g}^1 (387 cm^{-1}) and out-of-plane A_{1g} (404 cm^{-1}) modes, exhibit a separation of 17 cm^{-1} confirms the monolayer nature of the MoS_2 film. Similarly, Raman peaks located at approximately 249 cm^{-1} and 258 cm^{-1} are attributed to the in-plane vibrational mode (E_{2g}^1) and the longitudinal-acoustic mode (2LA (M)) of WSe_2, respectively. The presence of B_{2g} Raman peak at 310 cm^{-1}, indicative of interlayer interaction, confirms the multilayer nature of the as-grown WSe_2 film.*

Supplementary Figure 2



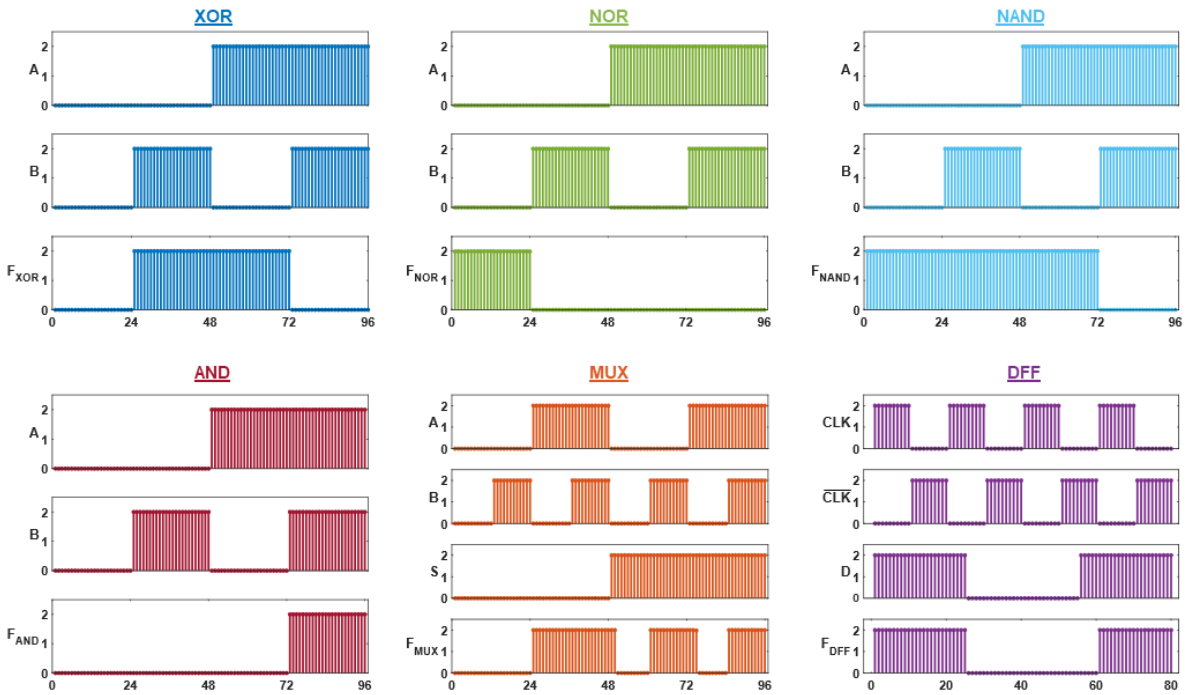
Supplementary Fig. 2. *Transfer characteristics of 20 WSe_2 field-effect transistors (FETs)* ***a)*** *before and* ***b)*** *after NO annealing treatment for 30 minutes at 100˚C.*

Supplementary Figure 3



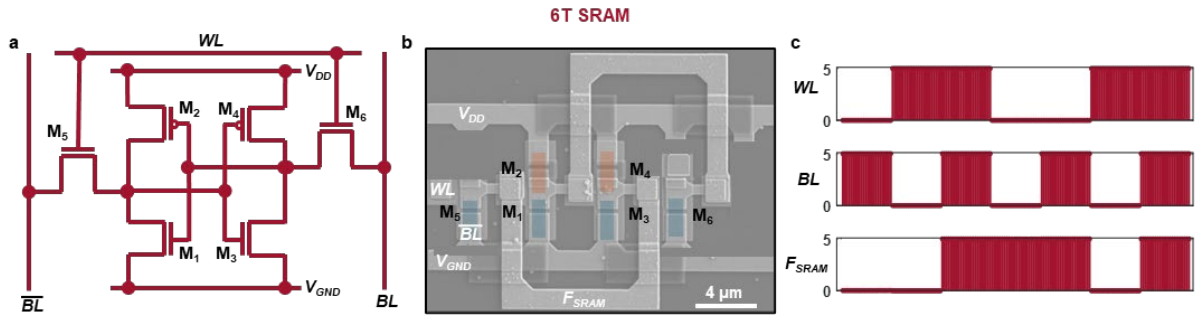
Supplementary Fig. 3. *Fabrication process flow for the 2D CMOS chip.*

Supplementary Figure 4



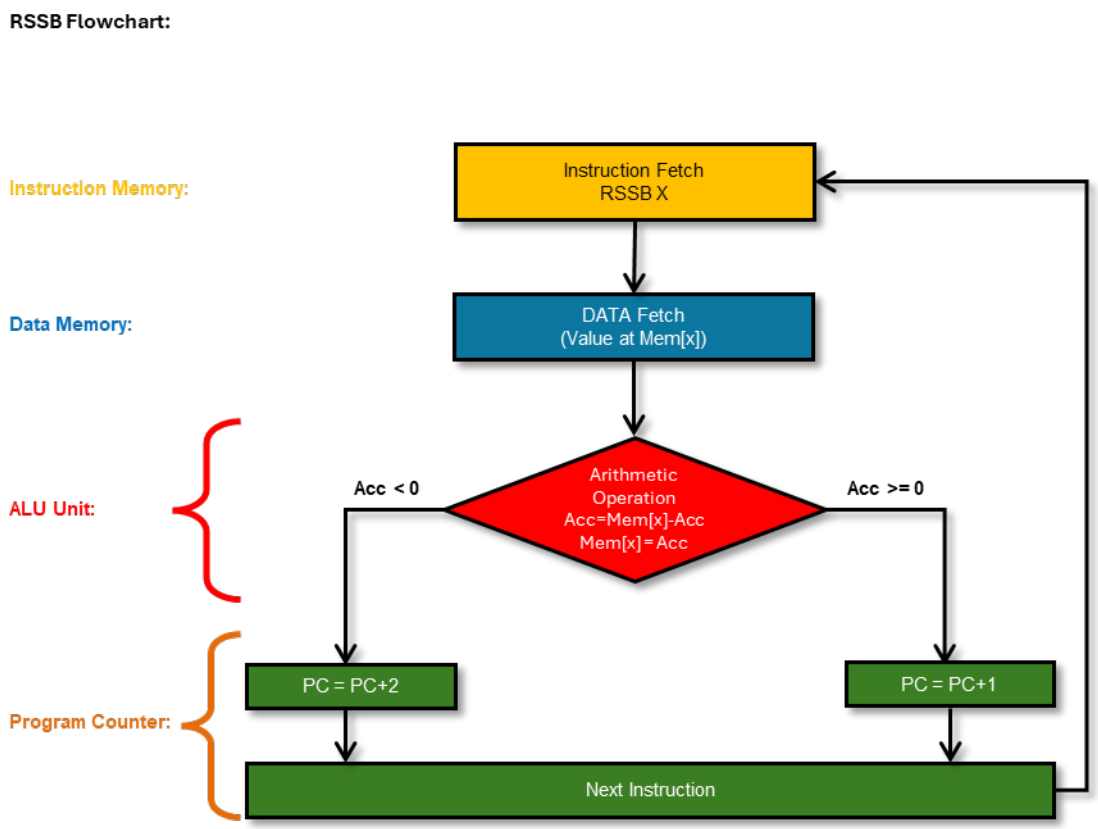
Supplementary Fig. 4. *Timing diagrams for logic gates showing logic operation at a supply voltage of 2 V.*

Supplementary Figure 5



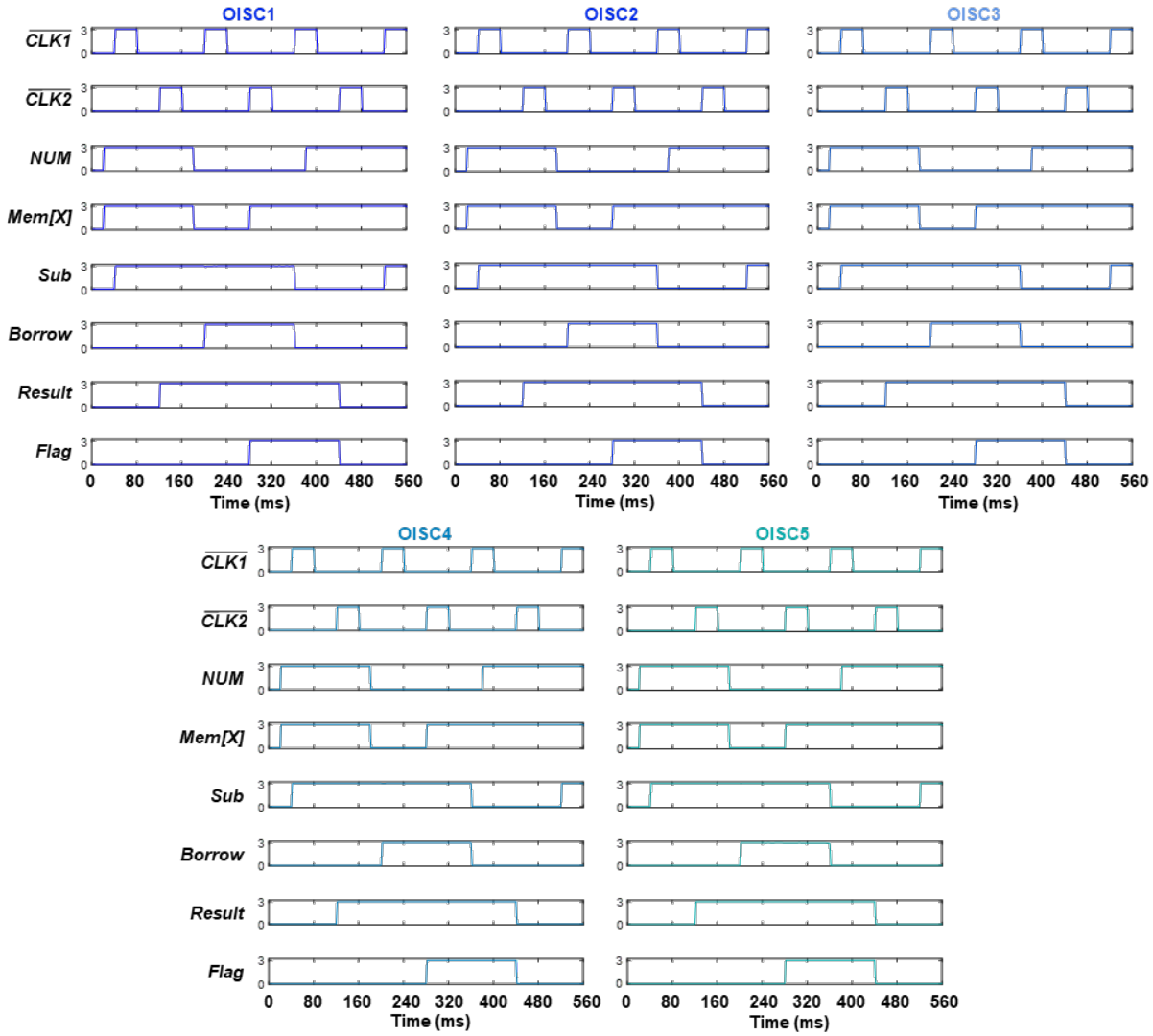
Supplementary Fig. 5. a-b) *Present the circuit diagram and False-colored SEM image of a 6T SRAM cell respectively, which consisting of six transistors (M_1–M_6). M_2 and M_4 are WSe_2-based PMOS transistors functioning as pull-up transistors, while M_1 and M_4 are MoS_2-based NMOS transistors acting as pull-down transistors. Together, they form two cross-coupled CMOS inverters, serving as a bistable latch to store binary data. M_2 and M_4 are connected to the power supply (V_{DD}) to maintain the memory cell in a high voltage state, whereas M_1 and M_3 are connected to ground (V_{GND}) to pull the cell to a low voltage state. The access transistors, M_5 and M_6, are MoS_2-based NMOS transistors essential for read/write operations. When word line (WL) is activated, they connect the bistable latch to the bit lines (BL & $\overline{BL}$), enabling data transfer.* ***c)*** *provides the timing diagram of WL, BL, and output F_{SRAM}, validating the proper operation 2D CMOS 6T SRAM cell.*

Supplementary Figure 6



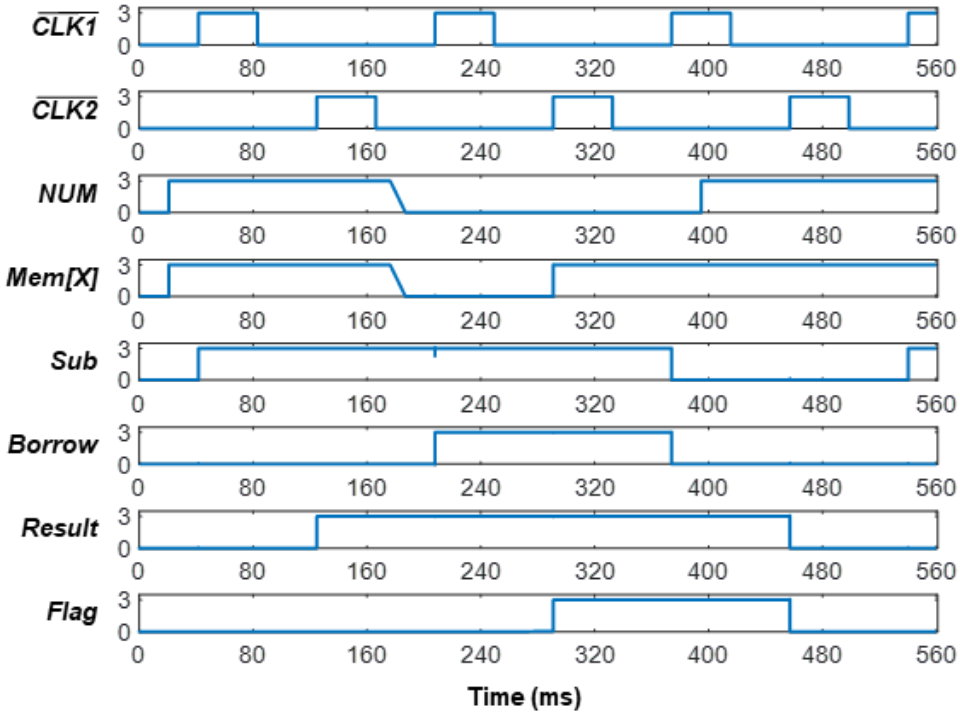
Supplementary Fig. 6. Flowchart for Reverse Subtract and Skip if Borrow (RSSB) Instruction: *Where in the first stage, instruction fetch, involves providing the instruction memory with the address to be read. At the first clock cycle (CLK1), the RSSB command is retrieved from the instruction memory and stored in a DFF. The RSSB instruction inherently includes the address for the subsequent instruction (e.g., RSSB X, where X denotes the address of the data). In the second stage, the processor reads the address for the data and fetches it from data memory. During CLK1, the data address is accessed, and its corresponding value is stored in another DFF. The third stage involves the ALU operation, which performs the necessary computations, such as subtracting the accumulator from the memory and comparing the result with zero. This computation is based on the data values supplied by the data-fetch unit and the accumulator. Finally, the write-back stage is responsible for saving the result of the RSSB operation back into the data memory at the address specified in the instruction. During the second clock cycle (CLK2), the outcome from the ALU is preserved in two DFFs. Concurrently, CLK2 activates the write-back process to the data memory, with the DFFs from the instruction-fetch unit providing the data address and the DFF from the write-back stage delivering the value to be written. This systematic approach ensures that each instruction is fetched, processed, and stored accurately, maintaining the integrity and efficiency of the processor's operations.*

Supplementary Figure 7



Supplementary Fig. 7. *RSSB operation on five additional 2D CMOS OISCs from the same chip.*

Supplementary Figure 8



Supplementary Fig. 8. *Simulated timing diagrams for the One Instruction Set Computer (OISC).*